

Amr Abed

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SUMMARY

Computer Engineer, Researcher, and Instructor. I got my Ph.D. in Computer Engineering from Virginia Tech in 2017. My current interests are at the intersection of machine learning and mobile and cloud computing and security. My Ph.D. research focused on applying machine learning to secure cloud containers through intrusion detection and remediation. Skilled, experienced, and passionate about software development using Java, Android SDK, C++, and C among others.

EDUCATION

Ph.D. Computer Engineering, August 2014 – July 2017

M.S. Computer Engineering, August 2010 – July 2013

Virginia Tech, Blacksburg, VA

GPA 3.8/4.0

B.S. Computer and System Engineering, September 2000 – June 2005

Ain-Shams University, Cairo, Egypt

Very Good (83.9%) with Honor, with Excellent project grade

TECHNICAL SKILLS

Java, Android, C++, C, Python, Kotlin, JavaScript

TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, Matplotlib

Docker, Firebase, Google Cloud Platform (GCP), Amazon Web Services (AWS)

SQL/SQLite, NoSQL, HTML, PHP

Linux Shell Scripting, LaTeX, MATLAB

LANGUAGES

Arabic, English, French

CURRENT ROLE

Founder, January 2020 – Present

Abed Solutions, Mississauga, ON, Canada

TECHNICAL EXPERIENCE

Learning Content Team Lead, December 2018 – August 2019

RoboGarden Inc, Cairo, Egypt

- Created content for a Python Machine Learning course to be taught in a continuing-education program of one of the top universities in Canada
- Led a team of 6 engineers to deliver contents for 6 different courses

Software Engineering Intern, May 2014 – August 2014

Google Fit Team, Google, Mountain View, CA

- Implemented an Android application that communicates with Google visual-search service and Google knowledge graph
- Collaborated with members from three different teams to successfully complete project on time
- Added features and fixed bugs for two other Android applications using the Google Fit APIs

Software Development Engineer, November 2007 – November 2008

Device Modeling Team, Mentor Graphics, Cairo, Egypt

- Studied and analyzed the C-based version of the Spectre2Eldo syntax converter
- Designed and implemented the OO-based version of the Spectre2Eldo syntax converter using C++ STL
- Solved problems related to using a Flex & Bison parser with C++

RESEARCH EXPERIENCE

Graduate Research Assistant, May 2016 – July 2017

Region Urban Living Lab, Virginia Tech, Arlington, VA

- Constructed a Raspberry Pi based IoT sensor for the MetroLab Network
- Designed and implemented a framework for the device to communicate with a remote MySQL server
- Implemented a website using PHP and Plotly.js to display data from the database

Graduate Research Assistant, January 2017 – May 2017

Hume Center, Virginia Tech, Arlington, VA

- Collaborated with members from Fujitsu in a joint research project to build an SDN-based security solution
- Designed the threat model for testing the security solution using Metasploit
- Implemented a container-based web application as an attack target

Graduate Research Assistant, January 2015 – May 2016

Hume Center, Virginia Tech, Blacksburg, VA

- Assisted in writing a winning proposal for S^2ERC , NSF Industry-University Cooperative Research Center
- Designed and implemented RHIDS, an intrusion detection system for cloud containers
- Tested the system using Docker containers deployed on Amazon-AWS EC2 instances

Graduate Research Assistant, February 2013 – May 2013

Magnum Group, Virginia Tech, Blacksburg, VA

- Implemented the Docker-EC2 tool using Shell scripting to automate running and testing network applications using a Docker container on different instances of the Amazon-AWS EC2
- Maintained and troubleshot build issues of the SODA-Cloud project (Mobile/Cloud Computing project funded by Siemens) on GitHub and BuildHive using Maven

TEACHING EXPERIENCE

Assistant Professor, November 2017 – January 2020

Minia University, Minia, Egypt

- Introduced and created 3 new undergraduate courses
- Taught 4 different courses to undergraduate students at 3 different levels
- Supervised a machine-learning based graduation project of a team of 6 undergraduate students

Instructor, March 2006 – August 2010

Inform - IBM Authorized Training Center, Cairo, Egypt

- Taught C and C++ Programming to more than 150 students from different backgrounds, many of them were from outside the Engineering field

Instructor, February 2008 – August 2010

Al-Sedeek Education Center, Giza, Egypt

- Taught C and C++ programming, and Algorithms & Data Structures using C and C++ courses to more than 300 students

Teaching Assistant, March 2007 – August 2010

Minia University, Minia, Egypt

- Taught 8 different undergraduate courses to 4 different classes of an average of 25 students each

VOLUNTEER WORK

IEEE Peer Reviewer, August 2016 – Present**Docker Mentor**, March 2017 – May 2018

SELECT PUBLICATIONS

Amr Abed. *Securing Cloud Containers through Intrusion Detection and Remediation*. Ph.D. dissertation, Virginia Tech, 2017.

Amr Abed, Mohamed Azab, Charles Clancy, and Mona Kashkoush. **Resilient Intrusion Detection System for Cloud Containers**. *International Journal of Communication Networks and Distributed Systems*, 2018.

Amr Abed, Charles Clancy, and David Levy. **Intrusion Detection System for Applications Using Linux Containers**. In *Security and Trust Management*, volume 9331 of *Lecture Notes in Computer Science (LNCS)*, pages 123–135, 2015.